

SANDY LAKE A, ON, Canada

WMO: 715870

Lat: 53.063N

Lon: 93.344W

Elev: 951

StdP: 14.20

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.2	-26.3	-38.8	0.6	-30.9	-33.5	0.9	-25.9	18.6	2.3	16.7	0.2	3.1	300	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	81.9	64.7	79.2	63.3	76.6	62.1	68.2	77.8	66.2	74.9	64.4	72.6	7.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.0	95.7	71.7	63.0	89.2	70.0	61.1	83.4	68.8	33.1	78.1	31.5	75.0	30.1	72.8	76.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.8	15.7	13.9	-37.9	89.9	4.0	3.9	-40.8	92.6	-43.1	94.9	-45.4	97.1	-48.3	99.9		
			-38.8	72.3	3.3	1.9	-41.2	73.7	-43.2	74.8	-45.0	75.9	-47.5	77.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	31.9	-2.7	-0.7	14.3	30.2	46.2	58.8	64.3	61.2	51.1	38.1	18.2	1.6	(6)
(7)		DBStd	26.42	14.00	12.89	13.31	12.75	9.89	7.50	6.35	6.40	8.84	8.77	12.96	14.37	(7)
(8)		HDD50	7902	1634	1420	1107	597	195	13	0	2	92	385	955	1502	(8)
(9)		HDD65	12226	2099	1840	1572	1045	586	210	91	152	426	833	1405	1967	(9)
(10)		CDD50	1294	0	0	0	2	77	279	443	351	124	17	1	0	(10)
(11)		CDD65	139	0	0	0	0	3	25	68	35	8	0	0	0	(11)
(12)		CDH74	1185	0	0	0	0	70	237	571	246	60	1	0	0	(12)
(13)		CDH80	202	0	0	0	0	11	23	124	34	10	0	0	0	(13)
(14)	Wind	WSAvg	6.4	6.1	6.0	6.7	6.6	7.2	6.6	6.2	6.1	6.9	6.6	5.6	(14)	
(15)	Precipitation	PrecAvg	21.8	0.9	0.7	0.9	1.0	1.8	2.7	3.6	3.3	2.6	2.0	1.3	0.9	(15)
(16)		PrecMax	28.9	2.0	1.5	2.0	2.0	3.5	5.4	5.7	6.5	4.5	5.8	2.0	2.0	(16)
(17)		PrecMin	18.0	0.3	0.2	0.5	0.1	0.6	1.1	1.5	1.1	0.6	0.6	0.6	0.3	(17)
(18)		PrecStd	2.7	0.4	0.3	0.4	0.5	0.7	0.9	1.0	1.2	1.1	1.0	0.4	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	35.4	55.2	64.5	81.1	82.8	89.1	84.5	79.9	68.0	52.5	35.0	(19)
(20)			MCWB	32.5	31.9	47.1	47.3	58.3	64.2	69.9	68.2	62.9	56.1	48.2	31.8	(20)
(21)		2%	DB	29.7	30.2	45.1	59.2	74.6	79.7	83.1	79.6	74.6	60.9	44.1	29.5	(21)
(22)			MCWB	29.0	28.1	37.5	45.1	55.2	62.2	66.4	65.0	63.0	52.3	41.6	28.0	(22)
(23)		5%	DB	24.9	24.9	38.2	54.6	68.8	76.8	80.0	76.6	69.9	56.1	38.3	26.2	(23)
(24)			MCWB	23.7	22.7	32.4	42.8	53.1	60.6	64.4	63.3	60.5	50.7	36.2	25.3	(24)
(25)		10%	DB	18.7	20.0	33.9	49.6	63.5	73.5	76.9	73.3	65.2	51.3	33.9	22.1	(25)
(26)			MCWB	17.8	18.2	30.8	39.9	51.6	59.2	62.9	62.4	57.6	47.3	32.6	21.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.8	33.6	46.2	49.4	62.2	69.6	72.3	69.6	67.8	57.6	47.4	32.2	(27)
(28)			MCD B	33.2	35.3	56.2	61.6	74.9	77.6	84.3	80.4	76.6	64.2	51.8	33.8	(28)
(29)		2%	WB	29.1	27.8	37.5	46.0	57.9	65.2	69.1	67.0	64.4	54.4	41.3	28.4	(29)
(30)			MCD B	29.5	29.7	43.5	57.9	69.1	73.9	78.8	75.7	72.5	59.5	43.7	29.4	(30)
(31)		5%	WB	23.7	22.8	33.3	43.3	54.9	63.1	66.6	65.3	61.2	50.7	36.3	25.4	(31)
(32)			MCD B	24.8	24.6	36.4	53.0	66.2	72.7	76.1	73.3	67.6	55.1	38.1	26.3	(32)
(33)		10%	WB	17.8	18.1	30.5	40.2	52.1	61.1	64.9	63.5	58.4	47.2	32.7	21.5	(33)
(34)			MCD B	18.6	20.1	34.6	48.9	62.9	70.5	73.4	71.2	65.1	51.7	33.9	22.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.5	20.6	23.3	23.2	21.3	20.1	19.5	18.2	16.4	11.5	11.6	13.6	(35)
(36)			MCDBR	16.4	21.6	23.9	27.5	28.8	24.6	23.5	22.7	22.0	19.2	14.4	12.1	(36)
(37)			MCWBR	15.8	19.0	17.6	16.2	14.3	11.2	10.3	11.0	12.1	12.3	11.6	11.0	(37)
(38)		5% WB	MCD BR	16.5	20.6	20.0	25.7	25.6	20.5	19.9	19.4	19.4	17.5	13.9	11.9	(38)
(39)			MCWBR	15.9	18.4	15.6	15.6	13.6	11.0	10.3	10.9	12.1	11.9	11.6	10.9	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.239	0.255	0.275	0.319	0.343	0.371	0.386	0.364	0.328	0.291	0.263	0.229	(40)
(41)		taud		2.285	2.280	2.318	2.392	2.455	2.396	2.392	2.453	2.524	2.585	2.426	2.293	(41)
(42)		Ebn at Noon		244	275	292	291	288	279	272	273	272	263	235	226	(42)
(43)		Edh at Noon		18	26	31	33	33	36	35	31	26	20	16	15	(43)
(44)	All-Sky Solar Radiation	RadAvg		334	713	1171	1649	1676	1759	1719	1410	969	539	269	210	(44)
(45)		RadStd		36	47	91	118	137	166	133	111	128	60	28	33	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page